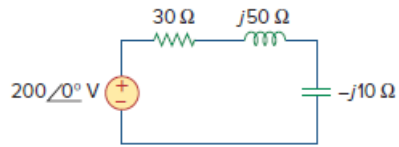


Problem

Given the circuit in Fig. 11.35, find the average power supplied or absorbed by each element.

Figure 11.35:



Step-by-step solution

Step 1 of 3

Refer to Figure 11.35 in the text book.

Calculate the current I flowing in the circuit.

$$I = \frac{V}{R + jX_L}$$

Substitute $200\angle 0^\circ$ V for V , $30\ \Omega$ for R , $-j10\ \Omega$ for jX_C and $j50\ \Omega$ for jX_L .

$$\begin{aligned} I &= \frac{200\angle 0^\circ}{30 + j50 - j10} \\ &= \frac{200\angle 0^\circ}{30 + j40} \\ &= \frac{200\angle 0^\circ}{50\angle 53.13^\circ} \\ &= 4\angle -53.13^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I flowing in the circuit is, $4\angle -53.13^\circ$ A.

Step 2 of 3

Calculate the average power P_R absorbed by the resistor.

$$P_R = \frac{1}{2} |I|^2 R$$

Substitute 4 A for $|I|$ and 30Ω for R .

$$\begin{aligned} P_R &= \frac{1}{2} (4)^2 (30) \\ &= (16)(15) \\ &= 240 \text{ W} \end{aligned}$$

Therefore, the value of the power P_R absorbed by the resistor is, 240 kW.

The value of the average power absorbed by the inductor is, 0 W.

The value of the average power absorbed by the capacitor is, 0 W.

[Comments \(3\)](#)

☐

Anonymous

why is 0W for inductor and capacitor?

☐

Anonymous

I'm not sure, but my best guess is as follows: When you apply the above equation with the resistance values from the inductor and capacitor, j terms will appear, making the power absorbed a Reactive Power on the Power Triangle. Therefore, the power absorbed is 0. Don't quote me on this to your linear circuits professor though, I'm just an amatuer

☐

Anonymous

cap and ind do not absorb average power

Step 3 of 3

Calculate the value of the average power P supplied by the voltage source.

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

Substitute 4 A for I_m , 200 V for V_m , 0° for θ_v and -53.13° for θ_i .

$$\begin{aligned} P &= \frac{1}{2} (200)(4) \cos(0^\circ - (-53.13)^\circ) \\ &= (400) \cos(53.13^\circ) \\ &= (400)(0.6) \\ &= 240 \text{ W} \end{aligned}$$

Therefore, the value of the average power P supplied by the voltage source is, 240 W.